

Temperature and Cardiovascular Admissions in Hong Kong

HKO exposure file complete; preparing the first HA merge

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READY: 132-month HKO exposure file **PENDING:** HA/HPC access and final schema analysis unit, lag plan, next deliverable **TODAY:**

What is ready for the first merge

Completed

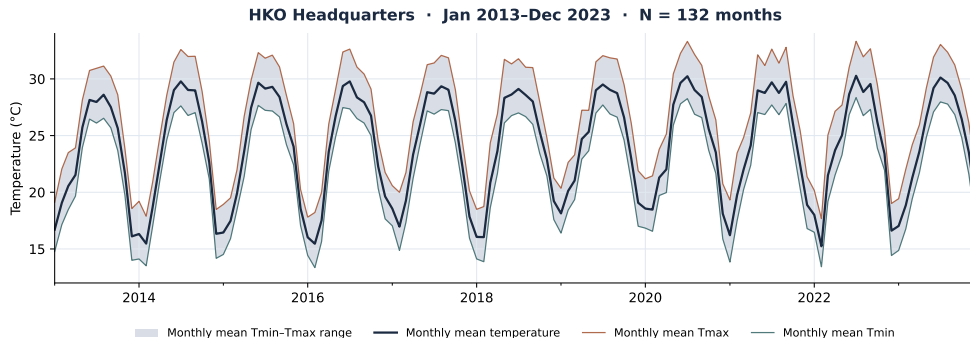
- HKO Headquarters monthly file, January 2013–December 2023 ($N = 132$)
- Monthly mean, $T_{\min}$, $T_{\max}$, and extreme counts
- Lag variables constructible
- Environmental portion of Table 1 completed
- Merge/model pipeline prepared

Confirmed by Roro

- ICD-9 coding
- Complete admission outcomes at patient level
- Ages 65–69 and 70–74 available separately
- ED-to-inpatient episodes are not included in the extract
- HPC onboarding follows wet-ink authorization

Current boundary: exposure file and merge pipeline are ready; real HA associations await HPC access and row-definition confirmation.

Monthly temperature at HKO Headquarters, 2013–2023



Environmental exposure data only; no admission outcomes shown.

Table 1 is partly complete

Panel A — Environmental observations

Unit: month $N = 132$ months HKO Headquarters

Variable	Mean	SD
Mean temperature (°C)	24.02	4.70
Mean Tmin (°C)	22.10	4.69
Mean Tmax (°C)	26.66	4.82
Hot nights (days/month)	3.40	5.40
Very hot days (days/month)	3.19	5.18
Cold days (days/month)	1.10	2.55

These variables can be described as requested; their model role depends on the at-risk population and pre-event timing.

Panel B — Patient/admission

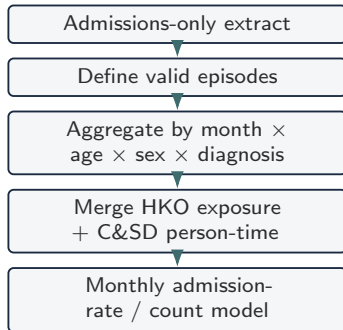
Unit and denominator pending confirmation of the HA schema.

- Patients and qualifying episodes
- Age and sex
- Medication fields: diuretics, beta blockers, metformin, SGLT2 inhibitors
- BMI and BMI missingness

One schema question determines the analysis unit

Does the extract contain admissions only, or a defined cohort with non-event follow-up?

Proposed default for the first analysis



If non-event follow-up exists:
patient-month risk model

Temperature variables are ready; here is the proposed lag plan



1. Analysis file can carry same-month and lag-1 mean, Tmin, and Tmax.
2. Same-month and lag-1 Tmax: $r \approx 0.82$ (131 pairs).
3. Fit temperature specifications separately rather than in one saturated model.
4. Proposed first sequence:
 - same-month exposure as initial specification
 - lag-1 exposure as sensitivity
 - current/previous-month average as secondary
5. Extend weather input to December 2012 so January 2013 has valid lag-1.

Decision for today: primary temperature metric — mean temperature, Tmin/Tmax, or extreme-day counts.

Decisions that unlock the first HA report

1. Outcome and episode definition

full meaning of DAE; exact row unit; principal versus any-position diagnosis; transfers and recurrent admissions; admission route/elective status.

2. Analysis population

admissions-only data or defined at-risk cohort; availability of medication dates, BMI dates, and non-event follow-up; primary temperature metric for the first model.

3. Access and governance

wet-ink/HPC timing; whether the expanded work requires an IRB amendment; permitted output and disclosure-review process.

Next deliverable after access:

validated HA data dictionary/QC report, descriptive admission counts, age-specific rates, and completed Table 1.

Regression will begin only after the outcome definition and descriptive QC are approved.

Appendix

Technical detail for discussion if needed

A1. Provisional ICD-9 definitions

Coding system **confirmed ICD-9** (Roro, 12 July). Code lists remain provisional until principal-diagnosis and local HA rules are confirmed.

Group	Codes	Notes
AMI	410.x	Exclude old MI 412
Hemorrhagic stroke	430, 431; provisionally 432.x	Sensitivity: 430–431 only
Ischemic stroke	433.x1, 434.x1, 436	Sensitivity: with/without 436
Exclude	412; 435.x; 438.x	Old MI; TIA; stroke sequelae

Source: `memos/data_request_roro.md`. ICD-10 backup retained only if needed historically.

A2. Full HKO validation table (annual extremes)

Station: HKO Headquarters. Result: all 33 annual extreme-day totals matched HKO *The Year's Weather*.

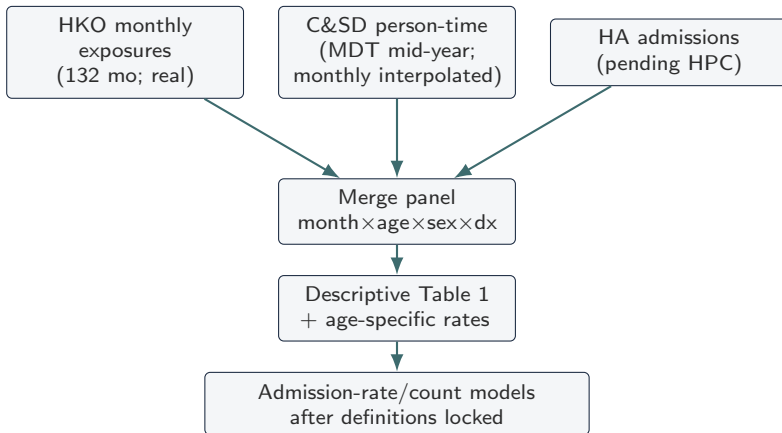
Year	Hot nights	Very hot days	Cold days
2013	10	17	14
2014	34	33	21
2015	37	28	7
2016	36	38	21
2017	41	29	9
2018	26	36	21
2019	46	33	1
2020	50	47	11
2021	61	54	13
2022	52	52	13
2023	56	54	14

Source: `outputs/tables/hko_annual_extremes_validation.csv`.

A3. Detailed HA schema questions

1. What does “DAE records included” mean in this ICD-9 extract? (No expansion invented.)
2. Is each row one admission episode, with drugs/BMI attached if available?
3. Can principal diagnosis be distinguished from secondary diagnoses?
4. Which admission-route or elective/emergency fields are available?
5. How are inter-hospital transfers and same-patient recurrence handled?
6. Are diuretics, beta blockers, metformin, SGLT2is, and BMI already on the approved extract?
7. What medication and BMI measurement dates are available relative to admission?
8. Does the extract contain non-event follow-up / a defined at-risk cohort, or admissions only?
9. What small-cell suppression rules apply to descriptive outputs from HPC?

A4. Pipeline / data flow



Scripts: 02.build_weather_monthly.R, 05_build_population_denominators.R, 08b.merge_real_ha_panel.R. Synthetic track exists only for workflow testing — not findings.

Population status: CSD_IMPORTED from Table 110-01001; monthly values are interpolated.

A5. Full Table 1 shell

Panel A — filled ($N = 132$ months; HKO Headquarters)

Mean temp 24.02 (4.70) °C; T_{min} 22.10 (4.69) °C; T_{max} 26.66 (4.82) °C; hot nights 3.40 (5.40); cold days 1.10 (2.55); very hot days 3.19 (5.18).

Panel B — pending HA access

Field	Status / note
N patients / episodes	Pending; denominator depends on schema
Age, sex	Requested; strata include 65–69 and 70–74
Diuretics; beta blockers	Requested descriptively; model role TBD
Metformin; SGLT2 inhibitors	Requested descriptively; model role TBD
BMI (+ missingness)	Requested descriptively; pre-event timing TBD

These covariates will be described as requested. Model use depends on cohort structure and measurement timing.

A6. Deferred pollution, district, and 2024 extensions

- **Pollution:** EPD import scripts are ready; the current file is a placeholder and will not be used for inference. Near-term priority remains temperature → AMI/stroke after HA access.
- **District / multi-station weather:** HKO Headquarters is the primary station. Multi-station or district extensions are optional sensitivities (Hogan), not required for the first HA report.
- **2024 weather extension:** Optional; raw 2024 extract exists. Do not delay the 2013–2023 HA core.